

33rd JK Conference Korean Workshop

Session 2

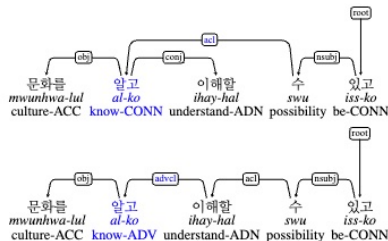
Hakyung Sung
University of Hawaii at Manoa

LCLB G20, UIUC

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Session 1 (Review)

- Overview of UD's goals and design
- Elements of UD annotations
- Korean UD corpora
- Annotation decisions and criteria
 - See our recent paper [Parser agreement/disagreement in L2 Korean UD](#)



Topics to cover

- Intro (5 mins)
- Annotate Korean texts (20 mins)
- Calculate frequencies (20 mins)
- Generate dependency-based concordance lines (20 mins)
- Conclusion (5 mins)
- Showcase how we do UD annotations and basic corpus analyses with a NLP tool

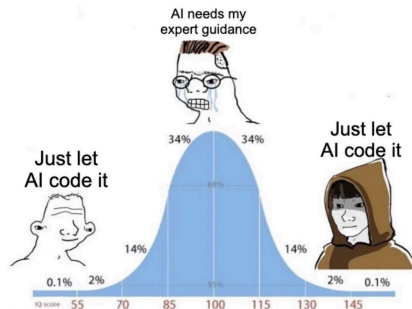
GitHub Repository: https://github.com/hksung/33JKConference_KoreanWorkshop

Colab Notebook: https://colab.research.google.com/drive/1RJdvKyT_bJPM6M0EFZ2acxUe-feehdCn?usp=sharing

** Please save a copy of Notebook to your Drive before starting the session.*

Colab notebook

- You do not need to write the Python code from scratch; the notebook already provides it (Simply click the ▶ button to run each code cell)
- To apply the workflow to your own research, you might need to **modify** parts of the code
- AI can be helpful; BUT still requires a minimal understanding of what the code is doing



1. Annotating Korean Texts

with *Stanza*

What we will do

- Install and import the required Python packages
- Download the Korean model
- Build a Korean NLP pipeline
- Load sample texts
- Generate UD annotations

1.1. Install and import the required packages

- Run the following cell in Colab:

```
!pip install stanza pandas
```

- **stanza** provides tools for tokenization, part-of-speech tagging, lemmatization, and dependency parsing
 - <https://stanfordnlp.github.io/stanza/>
- **pandas** allows us to organize annotations as tables/dataframes

What does `pip install` mean?

`pip install` downloads and installs Python packages that are not already available in the Colab *environment*.

What does `!` mean?

The exclamation mark tells Colab to run the line as a terminal command rather than as regular Python code.

1.1. Install and import the required packages

```
import re
from pathlib import Path
from IPython.display import display

import stanza
import pandas as pd
```

Why do we import packages?

Importing makes tools from built-in and newly installed Python libraries available in the notebook.

- re: finds/splits text patterns
- Path: works with file/folder paths
- display: displays tables neatly in Colab
- stanza, pandas

1.2. Download the Korean GSD model and build the pipeline

```
PROCESSORS = "tokenize,pos,lemma,depparse"
```

```
stanza.download(  
    lang="ko",  
    package="gsd",  
    processors=PROCESSORS,  
    verbose=False  
)  
  
nlp = stanza.Pipeline(  
    lang="ko",  
    package="gsd",  
    processors=PROCESSORS,  
    tokenize_no_ssplite=True,  
    use_gpu=False,  
    verbose=False  
)
```

What does this code do?

- Downloads Stanza's Korean GSD model
- Builds a pipeline, stored as `nlp`
- Applies four annotation steps:
 1. `tokenize`
 2. `pos`
 3. `lemma`
 4. `depparse`

1.3. Load and inspect two sample texts

```
import requests

BASE_URL = (
    "https://raw.githubusercontent.com/"
    "hksung/33JKConference_KoreanWorkshop/"
    "main/data"
)
```

What does this code do?

- requests: downloads files from a web address
- BASE_URL: stores the shared location of the sample files
- The files are downloaded directly from the workshop GitHub repository

1.3.2. Load and inspect two sample texts

```
def load_raw_sentences(filename):
    url = f"{BASE_URL}/{filename}"

    response = requests.get(url)
    response.raise_for_status()

    text = response.text.replace("\n", " ")

    sentences = [
        sentence.strip()
        for sentence in re.split(
            r"(?<=[.!?])\s+", text
        )
        if sentence.strip()
    ]

    return sentences
```

What does this function do?

- Receives a filename
- Downloads the corresponding text file
- Replaces line breaks with spaces
- Splits the text after sentence-final punctuation
- Returns a list of sentences

1.3.3. Load and inspect two sample texts

```
raw_sentences1 = load_raw_sentences("sample1.txt")
raw_sentences2 = load_raw_sentences("sample2.txt")

for sample_name, sentences in [
    ("Sample 1", raw_sentences1),
    ("Sample 2", raw_sentences2)
]:
    print(
        f"{sample_name} | "
        f"Number of sentences: {len(sentences)}"
    )

    for i, sentence in enumerate(sentences, start=1):
        print(f"{i}. {sentence}")

    print()
```

1.4.1. Apply UD-based annotation

```
def annotate_sentences(  
    raw_sentences,  
    document_label  
):  
    rows = []  
  
    for sentence_id, sentence_text \  
        in enumerate(raw_sentences, start=1):  
  
        doc = nlp(sentence_text)
```

What does this function do?

- Input: a list of sentences
- Assigns an ID to each sentence
- Applies the NLP pipeline to each sentence
- Creates rows to store the results

1.4.2. Apply UD-based annotation

```
doc = nlp(sentence_text)
for sent in doc.sentences:
    words_by_id = {
        word.id: word
        for word in sent.words
    }
    for word in sent.words:
        rows.append(
            { ...
              "text": word.text,
              "lemma": word.lemma,
              "upos": word.upos,
              "xpos": word.xpos,
              ...
            }
        )
```

Stanza output

Document → Sentence → Word

Word attributes (example)

- text: surface form, typically an eojeol (학생이)
- lemma: morphemes combined with + (학생+이)
- upos: Universal POS tag (NOUN)
- xpos: Korean-specific morpheme-level POS tags (NNG+JKS) ▶ [Sejong tag set](#)

1.4.3. Apply UD-based annotation

```
annotations1 = annotate_sentences(  
    raw_sentences1,  
    "text_1"  
)  
  
annotations2 = annotate_sentences(  
    raw_sentences2,  
    "text_2"  
)
```

What does this code do?

- Applies the same annotation function to both texts
- Stores the results in two separate tables

1.4.4. Select columns for the preview

```
preview_cols = [  
    "sentence",  
    "sentence_id",  
    "word_id",  
    "text",  
    "lemma",  
    "upos",  
    "xpos",  
]
```

What does this code do?

- Selects the columns to show in the annotation preview
- Stores the selected column names in `preview_cols`

1.4.5. Preview the annotations

```
print("Annotation for Text 1")
display(
    annotations1[preview_cols].head(20)
)

print("Annotation for Text 2")
display(
    annotations2[preview_cols].head(20)
)
```

What does this code do?

- Selects only the columns listed in `preview_cols`
- Displays the first 20 annotated words
- Allows us to inspect the annotations before continuing

Activity 1: Inspect the annotations

- Change the following values to inspect another sentence:
 - TEXT_TO_INSPECT = 1: select Text 1
 - SENTENCE_TO_INSPECT = 2: select Sentence 2
- Compare the upos and xpos tags [▶ Sejong tag set](#)
- Discuss any interesting and/or unexpected annotations with the person next to you.
- Keep in mind that these sentences are drawn from L2 Korean learner writing data [▶ L2KARG corpus](#)

1.5. Save the output into conllu files

```
...
def save_conllu(raw_sentences, text_label, output_path):
    output_path = Path(output_path)
    conllu_lines = []
    for sentence_id, sentence_text in enumerate(raw_sentences, start=1):
        doc = nlp(sentence_text)
        for stanza_sentence_id, sent in enumerate(doc.sentences, start=1):
            conllu_lines.append(
                f"# sent_id = {text_label}-{sentence_id}-{stanza_sentence_id}")
            conllu_lines.append(
                f"# text = {sent.text or sentence_text}")
            for word in sent.words:
                fields = [
                    word.id, word.text, word.lemma, word.upos, word.xpos,
                    word.feats, word.head, word.deprel, word.deps, word.misc]
                conllu_lines.append(
                    "\t".join(conllu_value(field) for field in fields))
            conllu_lines.append("")
    ...
```

1.5. Save the output into conllu files

```
save_conllu(  
    raw_sentences=raw_sentences1,  
    text_label="text1",  
    output_path="annotations1.conllu")
```

```
save_conllu(  
    raw_sentences=raw_sentences2,  
    text_label="text2",  
    output_path="annotations2.conllu")
```

Why save the annotations?

- Preserve the annotations after running the code
- Share/analyze them using other tools
- How can we download the files to a local computer from Colab?

2. Calculate frequencies

raw and normalized feature frequencies

What we will do

- Combine the two annotation dataframes
- Exclude punctuation and define normalization denominators
- Create a reusable frequency-calculation function
- Calculate raw and normalized UPOS frequencies
- Compare UPOS patterns across the two learner texts
- Calculate and visualize Korean-specific XPOS frequencies

2.1. Combine the annotation DataFrames

```
annotations = pd.concat(  
    [annotations1, annotations2],  
    ignore_index=True)  
  
...  
  
display(annotations.head(20))
```

What does this code do?

- Combines annotations1 and annotations2 into one DataFrame
- Stores the combined table as annotations

2.2. Define the normalization denominator

```
analysis_words = annotations.loc[
    annotations["upos"] != "PUNCT"
].copy()

word_counts_by_text = (
    analysis_words
    .groupby("document")
    .size()
    .reset_index(name="total_words")
)

if (word_counts_by_text["total_words"] == 0).any():
    raise ValueError(
        "At least one text has no "
        "non-punctuation syntactic words."
    )

print("Normalization denominators:")
display(word_counts_by_text)
```

Why normalize frequencies?

Raw frequencies are affected by text length. Normalization allows us to compare features across texts of different lengths.

What does this code do?

- Excludes punctuation from the frequency analysis
- Uses the remaining word counts as denominators

2.3. Create a reusable frequency function

```
def calculate_frequency(
    data, feature_column, denominator):

    if denominator == 0:
        raise ValueError(
            "The denominator is 0. ")

    frequency = (
        data[feature_column]
        .fillna("_")
        .value_counts()
        .rename_axis(feature_column)
        .reset_index(name="raw_frequency"))

    frequency["per_1,000_words"] = (
        frequency["raw_frequency"]
        / denominator
        * 1000
    )
    return frequency
```

Normalization formula

$$\frac{\text{raw frequency}}{\text{total words}} \times 1,000$$

What does this function do?

- Counts each value in a selected annotation column
- Stores the counts as raw_frequency
- Calculates normalized frequency per 1,000 words
- Returns the results as a new DataFrame

Activity 2. Calculate UPOS frequencies

```
upos_frequency_list = []

for document, data_subset \
    in analysis_words.groupby("document"):

    denominator = (
        TOTAL_WORDS_BY_TEXT[document]
    )

    frequency calculate_frequency(
        data=data_subset,
        feature_column="upos",
        denominator=denominator,
    )

    frequency["document"] = document
    upos_frequency_list.append(frequency)
```

```
upos_frequency = pd.concat(
    upos_frequency_list,
    ignore_index=True)

upos_frequency = upos_frequency[
    [
        "document",
        "upos",
        "raw_frequency",
        "per_1,000_words",
    ]]

upos_frequency = (
    upos_frequency
    .sort_values(
        by=["document", "upos"]
    )
    .reset_index(drop=True))

display(upos_frequency)
```

Output columns

- document: Text 1 or Text 2
- upos
- raw_frequency
- per_1,000_words: normalized frequency

Discussion questions

Text information

Text 1 was written by a lower-proficiency learner, whereas Text 2 was written by a higher-proficiency learner.

1. (BEFORE LOOKING AT THE RESULT!)
What preliminary hypothesis can you draw about the two learners' use of UPOS categories, and why?
2. If you only consider `raw_frequency`, which text appears to use more nouns, verbs, adjectives, and/or adverbs?
3. Does the interpretation change when you compare `normalized_frequency`?

Activity 3. Calculate XPOS frequencies

- XPOS provides detailed Korean-specific part-of-speech information
- One syntactic word may contain multiple XPOS tags
 - VV+ETM $\rightarrow$ VV and ETM
- We should first split these tag sequences before calculating frequencies.

```
xpos_rows = []

for _, row in analysis_words.iterrows():
    xpos_tags = str(
        row["xpos"]
    ).split("+") # SPLIT!

    for xpos_tag in xpos_tags:
        xpos_rows.append(
            {
                "document": row["document"],
                "xpos": xpos_tag,
            }
        )

xpos_long = pd.DataFrame(xpos_rows)
```

```

TOTAL_XPOS_BY_TEXT = (
    xpos_long
    .groupby("document")
    .size()
    .to_dict())

xpos_frequency_list = []

for document, data_subset \
    in xpos_long.groupby("document"):

    denominator = (TOTAL_XPOS_BY_TEXT[document])

    frequency = calculate_frequency(
        data=data_subset,
        feature_column="xpos",
        denominator=denominator,)

    frequency["document"] = document
    xpos_frequency_list.append(frequency)

```

What does this code do?

- The denominator is the total number of individual XPOS tags (i.e., XPOS frequency is normalized per 1,000 morphemes)
- The same reusable frequency function from the UPOS frequency calculation is applied

```
xpos_frequency = pd.concat(  
    xpos_frequency_list,  
    ignore_index=True)  
  
...  
  
xpos_frequency = xpos_frequency.rename(  
    columns={  
        "per_1,000_words":  
        "per_1,000_xpos_tags"})  
  
xpos_frequency = (  
    xpos_frequency  
    .sort_values(  
        by=["document", "xpos"])  
    .reset_index(drop=True))  
  
display(xpos_frequency)
```

Repeated from Activity 2

Combining, reordering, sorting, and displaying the results follow the same workflow as the UPOS analysis.


```
import matplotlib.pyplot as plt

TOP_N = 15

xpos_comparison = (
    xpos_frequency
    .pivot(
        index="xpos",
        columns="document",
        values="per_1,000_xpos_tags")
    .fillna(0))

xpos_comparison["total"] = (xpos_comparison.sum(axis=1))

top_xpos_comparison = (
    xpos_comparison
    .sort_values(
        "total",
        ascending=False)
    .head(TOP_N)
    .drop(columns="total"))
```

What does this code do?

- We have more XPOS tags!
- Reshapes the table for a plot
- Selects the 15 most frequent XPOS tags across both texts

You may modify

Change TOP_N to display more/fewer tags.

```

top_xpos_comparison = (
    top_xpos_comparison.rename(
        columns={
            "text_1": "Text 1 (lower proficiency)",
            "text_2": "Text 2 (higher proficiency)",})})

...

top_xpos_comparison = (
    top_xpos_comparison.loc[plot_order])

ax = top_xpos_comparison.plot(
    kind="barh",
    figsize=(10, 8))

ax.set_xlabel("Frequency per 1,000 XPOS tags")
ax.set_ylabel("XPOS tag")
ax.legend(title="Text")
plt.tight_layout()
plt.show()

```

Discussion questions

1. Which Korean-specific XPOS tags occur most frequently in general?
2. Which tags seem to distinguish the two proficiency levels, and what linguistic factors might explain the differences?
3. Beyond characterizing proficiency, how could XPOS annotations support research questions in your own area of interest?

Extra activity: Try new texts

- If you have extra time, return to previous sections and explore `sample3.txt` and `sample4.txt`
- Repeat the same workflow:
 - annotate the texts
 - inspect the POS tags
 - calculate normalized frequencies
 - visualize the XPOS patterns
- Samples 3 and 4 respond to a different prompt: *Should children start learning a foreign language at an early age?*

3. Generate dependency-based concordance lines

explore linguistic features in context

Concordance lines

- Frequency analysis tells us **how often** a linguistic feature occurs.
- However, frequency alone does not show **how** the feature is used **in context**.
- **Concordance lines** allow us to inspect recurring linguistic patterns in their original sentences.

What we will do

- Add head-word information to the dependency annotations
- **Select a dependency relation** ▶ UD tags
- Generate dependent-head concordance lines for the selected relation
- Select a dependent keyword and inspect the predicates it attaches to
- Compare predicate contexts and morphosyntactic patterns across texts

3.1. Load and annotate two new texts

```
raw_sentences5 = load_raw_sentences(  
    "sample5.txt"  
)  
  
raw_sentences6 = load_raw_sentences(  
    "sample6.txt"  
)  
  
annotations5 = annotate_sentences(  
    raw_sentences5,  
    "text_5"  
)  
  
annotations6 = annotate_sentences(  
    raw_sentences6,  
    "text_6"  
)
```

What does this code do?

- Loads Samples 5 and 6 from the workshop repository
- Reuses the functions defined in Section 1
- Applies the Korean annotation pipeline to both texts
- Stores the results as annotations5 and annotations6
- Will take about a minute!

3.1. Load and annotate two new texts (Check)

```
print(
    f"Text 5: "
    f"{len(raw_sentences5)} "
    f"input sentences"
)

print(
    f"Annotated syntactic words: "
    f"{len(annotations5):,}"
)

print(
    f"\nText 6: "
    f"{len(raw_sentences6)} "
    f"input sentences"
)

print(
    f"Annotated syntactic words: "
    f"{len(annotations6):,}"
)
```

3.2. Combine the annotation tables

```
annotations56 = pd.concat(  
    [annotations5, annotations6],  
    ignore_index=True  
)  
  
print(  
    f"\nTotal annotated syntactic words: "  
    f"{len(annotations56):,}"  
)  
  
display(annotations56.head(20))
```

- Q. What can you see?
- The head lemma shows the form of the head word, but not its morpheme annotations, which is the information I'm interested in.

3.3. Add the XPOS tag of each word's syntactic head

```
def add_head_xpos(data):
    output_rows = []
    for _, sentence_df in data.groupby(
        ["document", "sentence_id"],
        dropna=False, sort=False,):
        words_by_id = (sentence_df.set_index("word_id")
                       .to_dict("index"))
        for _, row in sentence_df.iterrows():
            head_id = row["head"]
            if head_id == 0:
                head_xpos = "ROOT"
            else:
                head_word = words_by_id.get(head_id)
                if head_word is None:
                    head_xpos = "_"
                else:
                    head_xpos = (head_word["xpos"])
            row_dict = row.to_dict()
            row_dict["head_xpos"] = head_xpos
            output_rows.append(row_dict)
    return pd.DataFrame(output_rows)
```

What does this function do?

- Retrieves the head word's Korean-specific POS tag
- Adds this information as a new head_xpos column

3.3. Add the XPOS tag of each word's syntactic head

```
annotations56_with_heads = (  
    add_head_xpos(annotations56)  
)  
  
head_preview_columns = [  
    ...  
    "head_upos",  
    "head_xpos",  
]  
  
display(  
    annotations56_with_heads[  
        head_preview_columns  
    ].head(20)  
)
```

3.4. Select a dependency relation

```
TARGET_RELATION = "nsubj" #subject
```

Other options

- obj: object
- obl: oblique nominal
- advcl: adverbial clause
- advmod: adverbial modifier
- nmod: nominal modifier

What does this code do?

Selects the dependency relation that we want to investigate.

Example

With `nsubj`, we can examine which subjects attach to which heads

3.4.1. Generate concordance lines

```
concordance = annotations56_with_heads.loc[
    annotations56_with_heads["deprel"]
    == TARGET_RELATION,
    [
        "sentence",
        "document",
        "sentence_id",
        "word_id",
        "text",
        "lemma",
        "upos",
        "xpos",
        "head",
        "deprel",
        "head_text",
        "head_lemma",
        "head_upos",
        "head_xpos",
    ],
].copy()
```

What does this code do?

- Keeps only rows with the selected relation
- Retains information about the dependent, head, and sentence

3.4.1. Generate concordance lines

```
concordance = concordance.rename(
    columns={
        "text": "dependent",
        "lemma": "dependent_lemma",
        "upos": "dependent_upos",
        "xpos": "dependent_xpos",
        "deprel": "relation",})

concordance = (
    concordance
    .sort_values(
        [
            "document",
            "dependent_lemma",
            "head_lemma",
            "sentence_id",])
    .reset_index(drop=True)
)

print(f"Target relation: {TARGET_RELATION}")
print(f"Number of concordance lines: "
      f"{len(concordance):,}")
```

What does this code do?

Renames the dependent columns, sorts similar patterns together, and displays the concordance *table*.

One row represents

dependent $\xrightarrow{\text{relation}}$ head

3.4.2. Identify frequent dependents

```
dependent_patterns = (
    concordance
    .groupby([
        "document",
        "dependent_lemma",
        "dependent_upos",
        "dependent_xpos",
    ], dropna=False,)
    .size()
    .reset_index(name="frequency")
    .sort_values([
        "document",
        "frequency",
        "dependent_lemma",
    ],
        ascending=[True, False, True],)
    .reset_index(drop=True))

display(dependent_patterns.head(30))
```

What does this code do?

Counts which dependent lemmas occur most frequently with the selected relation in each text.

3.4.3. Compare dependent lemmas

```
dependent_pattern_comparison = (  
    dependent_patterns  
    .pivot_table(  
        index=[  
            "dependent_lemma",  
            "dependent_upos",  
            "dependent_xpos",  
        ],  
        columns="document",  
        values="frequency",  
        fill_value=0,  
        aggfunc="sum",  
    )  
    .reset_index()  
)  
  
dependent_pattern_comparison[  
    "total"  
] = (  
    dependent_pattern_comparison["text_5"]  
    + dependent_pattern_comparison["text_6"]  
)
```

What does this code do?

Places the dependent frequencies from Texts 5 and 6 side by side for comparison.

3.4.3. Compare dependent lemmas

```
dependent_pattern_comparison = (  
    dependent_pattern_comparison  
    .sort_values(  
        "total",  
        ascending=False  
    )  
    .reset_index(drop=True)  
)  
  
display(  
    dependent_pattern_comparison  
    .drop(columns="total")  
    .head(20)  
)
```

What does this code do?

Sorts dependent lemmas by their combined frequency and displays the top 20.

	dependent_lemma	dependent_upos	dependent_xpos	text_5	text_6
0	수	NOUN	NNB	172	147
1	학생+들+이	NOUN	NNG+XSN+JKS	85	89
2	학생+들+은	NOUN	NNG+XSN+JX	64	58
3	인공+지능+이	NOUN	NNG+NNG+JKS	45	39
4	문장+이	NOUN	NNG+JKS	11	12
5	수업	NOUN	NNG	11	11
6	교수자+가	NOUN	NNG+JKS	10	11
7	학습자+는	NOUN	NNG+JX	15	5
8	학습자+들+이	NOUN	NNG+XSN+JKS	10	10
9	대학+생+들+이	NOUN	NNG+NNG+XSN+JKS	10	10
10	한국어	NOUN	NNG	10	8
11	인공+지능+은	NOUN	NNG+NNG+JX	8	8
12	위험+이	NOUN	NNG+JKS	0	16
13	학습	NOUN	NNG	0	15
14	도구+가	NOUN	NNG+JKS	1	13

3.4.4. Select a dependent keyword

```
TARGET_DEPENDENT_KEYWORD = (  
    dependent_pattern_comparison  
        .iloc[0]["dependent_lemma"]  
        .split("+")[0]  
)  
  
TARGET_DEPENDENT_KEYWORD = "학생"
```

What does this code do?

Selects one dependent keyword for closer inspection.

Why use a keyword?

학생 can match lemmas such as 학생+이, 학생+은, and 학생+들+이, which all store in one eojeol.

3.4.5. Add compact predicate context

- The syntactic head alone may not capture the complete Korean predicate

Review the result from 3.4.1. (e.g., 한국어(nsubj)-확인하지(VERB))

- So here we first locate the head and check up extra information:
 - auxiliary predicates (e.g., 있다, 되다)
 - bridge nouns (e.g., 수, 것)
 - negative expressions (e.g., 않다, 못하다, 없다)
- Relevant tokens are added to create a compact predicate context

3.4.5. What does the function create?

Predicate information

- `predicate_context`: surface forms
- `predicate_context_lemmas`: lemma sequence
- `predicate_context_xpos`: XPOS sequence
- `polarity`: affirmative-like, negative-like, root, or unknown

Morphosyntactic pattern

The function combines the dependent XPOS, dependency relation, and predicate XPOS:

$$\text{dependent XPOS} \xrightarrow{\text{relation}} \text{predicate XPOS}$$

e.g.,

$$\text{NNG+JKS} \xrightarrow{\text{nsubj}} \text{VV+EC} + \text{VX+EF}$$

3.4.5. Example predicate contexts

- 비교하다 head predicate only
- 비교하지 않다 predicate + negation
- 경험할 수 있다 predicate + bridge noun + auxiliary
- 배우게 되다 predicate + auxiliary continuation
- The function uses a simple rule-based heuristic!

3.4.6. Inspect the selected dependent keyword

```
selected_dependent_examples = (  
    concordance.loc[  
        concordance["dependent_lemma"]  
        .fillna("")  
        .str.contains(  
            TARGET_DEPENDENT_KEYWORD,  
            regex=False,  
        )  
    ]  
    .sort_values(  
        [  
            "document",  
            "dependent_lemma",  
            "head_lemma",  
            "sentence_id",  
        ]  
    )  
    .reset_index(drop=True)  
)
```

What does this code do?

Keeps only concordance lines
whose dependent lemma
contains the selected keyword.

3.4.6. Inspect the selected dependent keyword

```
selected_dependent_examples = (  
    add_predicate_context(  
        concordance_data=  
            selected_dependent_examples,  
            full_data=  
                annotations56_with_heads,))  
  
print(f"Target relation: "  
      f"{TARGET_RELATION}")  
  
print(f"Selected dependent keyword: "  
      f"{TARGET_DEPENDENT_KEYWORD}")  
  
print(f"Number of matching lines: "  
      f"{len(selected_dependent_examples):,}")  
  
display(  
    selected_dependent_examples[  
        selected_example_columns  
    ]  
)
```

What does this code do?
Adds predicate-context
information and displays all
matching concordance lines.

3.4.7. Compare predicate contexts

```
predicate_context_patterns = (  
    selected_dependent_examples  
        .groupby(  
            [  
                "document",  
                "predicate_context_lemmas",  
                "predicate_context",  
                "predicate_context_xpos",  
                "polarity",  
            ],  
            dropna=False,  
        )  
        .size()  
        .reset_index(name="frequency")  
)
```

What does this code do?

Counts which predicate contexts occur with the selected dependent in each text.

3.4.7. Compare predicate contexts

```
predicate_context_comparison = (  
    predicate_context_patterns  
        .pivot_table(  
            index=[  
                "predicate_context_lemmas",  
                "predicate_context",  
                "predicate_context_xpos",  
                "polarity",],  
            columns="document",  
            values="frequency",  
            fill_value=0,  
            aggfunc="sum",)  
        .reset_index()  
)  
  
predicate_context_comparison[  
    "total_frequency"  
] = (predicate_context_comparison[  
    document_columns  
] .sum(axis=1)  
)
```

What does this code do?

Compares predicate contexts across Texts 5 and 6 and calculates their combined frequency.

3.4.8. Compare morphosyntactic patterns

```
morphosyntactic_patterns = (  
    selected_dependent_examples  
    .groupby(  
        [  
            "document",  
            "morphosyntactic_pattern",  
        ],  
        dropna=False,  
    )  
    .size()  
    .reset_index(name="frequency")  
)
```

What does this code do?

Counts each dependent-predicate XPOS pattern in each text.

Example

$$\text{NNG+JKS} \xrightarrow{\text{nsubj}} \text{VV+EC} + \text{VX+EC}$$

3.4.8. Compare morphosyntactic patterns

```
morphosyntactic_comparison = (  
    morphosyntactic_patterns  
    .pivot_table(  
        index="morphosyntactic_pattern",  
        columns="document",  
        values="frequency",  
        fill_value=0,  
        aggfunc="sum",  
    )  
    .reset_index()  
)  
  
morphosyntactic_comparison["total"] = (  
    morphosyntactic_comparison["text_5"]  
    + morphosyntactic_comparison["text_6"]  
)  
  
display(  
    morphosyntactic_comparison  
    .drop(columns="total")  
)
```

What does this code do?

Places the pattern frequencies
from Texts 5 and 6 side by side.

3.4.9. Visualize the patterns

```
TOP_N = 10

plot_data = (
    morphosyntactic_comparison
    .head(TOP_N)
    .sort_values(
        "total",
        ascending=True)
    .set_index(
        "morphosyntactic_pattern"))

ax = plot_data[
    ["text_5", "text_6"]
].plot(
    kind="barh",
    figsize=(10, 6),
)

plt.tight_layout()
plt.show()
```

What does this code do?

Visualizes the ten most frequent morphosyntactic patterns across the two texts.

Activity 4. Interpret the concordance lines

Please go back to the codes and modify them based on the dependency relation and keyword you are interested in.

1. Which dependency relation are you analyzing? (e.g., nsubj, obj, obl...)
2. Which dependent lemma did you examine?
3. Which morphosyntactic patterns occur most frequently? Any interesting finding?
4. Are the recurring patterns similar or different across Texts 5 and 6?
5. Does this result show meaningful linguistic patterns?
 - If yes, how would you interpret it?
 - If no, what additional information would you add to the concordance line? How would you like to edit/rewrite the codes to do that?

Conclusion

Evaluation on L2 Korean

- Current NLP tools achieve overall high accuracy in morphosyntactic tagging and dependency parsing of L2 Korean.
- Note that their performance varies across different tags.

	Metric	Stanza	Trankit
Morpheme {	LEMMA	95.64	88.84
	XPOS	89.72	91.81
Dependency {	UAS	85.53	92.28
	LAS	80.36	89.13

Table 2: Performance comparison (F1 scores) of fine-tuned Stanza and Trankit models on the L2K-UD test dataset

Sung & Shin (2026). Parser Agreement and Disagreement in L2 Korean Universal Dependencies

Application: Korean Lexical Complexity Analyzer

- The Korean Lexical Complexity Analyzer automatically measures multiple dimensions of lexical complexity in Korean texts.
- It uses UD annotations to identify target lexical items.

Korean Lexical Complexity Analyzer (Demo)

Enter two Korean texts to compare shared indices.

Note. This is a lightweight demo (trimmed DB: freq > 10; default Stanza), so results may differ from the full package. For details, see the [repository](#) (Python package and index descriptions included).

Text 1

내가 협동 더 선호합니다.
협동 있으면 많이 사람들을 좋아하는 것은 받았습니다.
나는 사람과 함께 일하고 좋아요.
같이 우리는 더 좋아하는 만큼이에요.
경쟁 있으면 다 사람이 나쁜 경쟁 있어요.
이 경쟁 내가 싫어요.
우리는 다 사람이 세상에서 협동 있으면 그름 이 세상 더 좋네요.
그래서 내가 협동을 더 선호합니다.

Text 2

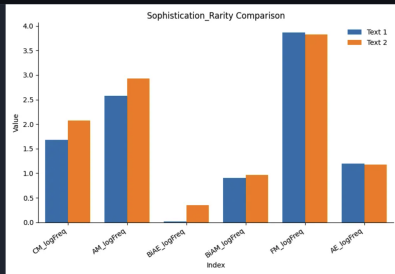
나는 협동을 더 선호하는 편이다.
그 이유는 협동을 하면 아이디어가 많아지고 혼자서는 떠오르지도 않았던 것을 할 수 있다고 생각하기 때문이다.
유학생활 때 그룹활동이 많았다.
이런 수업에서는 그룹원리를 했다.
한 동복 이미지마나 생각이 있어서 그것들을 정리하는건 결코 쉽지는 않았지만 마지막으로는 하나의 작품이 되어서 좋은 발표가 되
었다고 생각한다.
또 협동을 해서 부담을 줄일 수도 있다.
예를 들어 1000개 만들어야 하는 게 있다고 한다.
한 명으로 하는 건 힘들지만 10명으로 하면 한 명은 100개 만들면 된다.
부담뿐만 아니라 시간도 줄일 수 있다.
게다가 협동을 함으로써 사람들이 서로 웃어줄 수 있다는 이미지가 있다.
협동이라는 단어를 들었을 때 어감을 때도 있지만 전체적으로 긍정적인 이미지가 더 크다.

Index Group

Sophistication_Rarity

Compare

Comparison Plot



Demo: huggingface.co/spaces/hksung/KLC-demo

GitHub: github.com/hksung/Korean-lexical-complexity-analyzer

Thank you!

Contact: `sungh@hawaii.edu`

Appendix

Sejong tag set

- The Sejong tag set provides Korean-specific part-of-speech labels at the morpheme level
- It distinguishes fine-grained categories such as common nouns, case particles, verb endings, and auxiliary predicates
- See the full tag set and descriptions:
<https://nlpx12korean.github.io/UD-KSL/xpos/>

UD tags

- See the full list of dependency relations and examples (*tailored to Korean*):
<https://nlpx12korean.github.io/UD-KSL/ud/>

L2K-ARG corpus

- 484 argumentative essays written by L2 Korean learners
- Provides C-test scores and selected human-rated writing scores
- Can be accessed here: <https://github.com/NLPxL2Korean/L2K-ARG>